

Thermaterial: Program Ambient Heat Transfer Behaviors on Objects through Fluidic Composites

Tianyu Yu
Tsinghua University
Beijing, China
yty21@mails.tsinghua.edu.cn

Qingyu Hu
Tsinghua University
Beijing, China
huqy22@mails.tsinghua.edu.cn

Yige Fan
Shanghai Jiao Tong University
Shanghai, China
fan0402@sjtu.edu.cn

Weiye Xu
Tsinghua University
Beijing, China
xuwy20@mails.tsinghua.edu.cn

Zhixiang Zhang
Tsinghua University
Beijing, China
zhixiang19@mails.tsinghua.edu.cn

Haipeng Mi*
Tsinghua University
Beijing, China
mhp@tsinghua.edu.cn

Stefanie Mueller
MIT CSAIL
Cambridge, MA, USA
stefanie.mueller@mit.edu

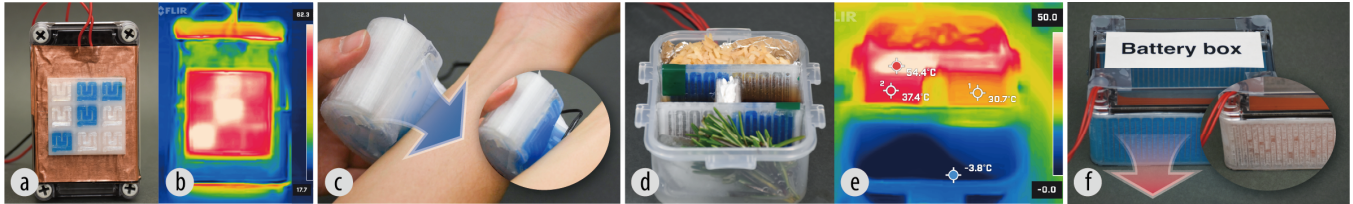


Figure 1: Programmable ambient heat transfer behaviors on objects using Thermaterial. (a,b)Passive thermal markers with reconfigurable thermal patterns. (c)Tunable thermal wearable with customized skin-contacting cooling temperature. (d,e)Containers with customized thermal-insulated layout. (f)Dynamic thermal-insulated enclosure for dynamic heat dissipation management.

ABSTRACT

Thermal interfaces in HCI have been widely explored. Previous work either used electronics that continuously consume energy to actively control the temperatures, or relied on materials that passively create only one predefined temperature behavior, which both leave the gap of exploring objects that allow self-contained programmable temperature behaviors, utilizing materials' dynamic thermal properties.

To address this gap, we introduce *Thermaterial*, a method using fluidic composites with controllable thermal properties to enable temperature behaviors without continuously consuming energy, instead utilizing programmed ambient heat transfer behaviors. We first presented the fluidic composition and fabrication method of Thermaterial and evaluated the tunable thermal insulation, revealing changes of up to 3.0 times. We then explored potential applications of objects with programmable ambient heat transfer behaviors,

*denotes the corresponding author.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3650747>

utilizing proof-of-concept prototypes built on Thermaterial. These include reconfigurable passive thermal markers, tunable thermal wearables, containers with customized thermal-insulated layouts, and dynamic thermal-insulated enclosures.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools.**

KEYWORDS

Programmable material, thermal interface, digital fabrication

ACM Reference Format:

Tianyu Yu, Yige Fan, Zhixiang Zhang, Qingyu Hu, Weiye Xu, Haipeng Mi, and Stefanie Mueller. 2024. Thermaterial: Program Ambient Heat Transfer Behaviors on Objects through Fluidic Composites. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3613905.3650747>

1 INTRODUCTION

Programmable materials have gathered extensive attention in the field of Human-Computer Interaction (HCI) due to their ability to create objects with interactive physical properties [11]. Previous works have studied programmable physical properties of

color [13], shape [3], texture [10], as well as optical [39] and magnetic properties [29]. However, limited attention has been given to programmable thermal properties of materials.

Interfaces with controllable temperature have a long tradition in the field of Human-Computer Interaction (HCI) for diverse purposes, such as on-body thermal feedback devices [8, 32], objects that display thermal information [17, 18], or thermoreactive tangible interfaces, e.g., thermochromic or shape-changing interfaces [24, 38]. However, most existing work relies on active heat generation and management devices for temperature control, such as using thermoelectric elements [31] or electrical fluidic systems [38], which have been limited in use, partly because of their continuous energy consumption and bulky connected hardware.

To address this limitation, researchers have explored how to utilize materials' intrinsic properties and their interaction with ambient heat transfer to create controllable temperature behaviors, which are fully passive and self-contained. For instance, Alex et al. [23] combined different thermally conductive materials on enclosures of devices to alleviate the overheat caused by heat concentration, and Zhang et al. [40] employed Voronoi air-exposure patterns on casts with localized density to regulate the local heat dissipation intensity. However, because these materials' thermal properties are static once fabricated, the object can only create one predefined heat transfer behavior during use. Objects comprising materials with controllable thermal properties, which allow self-contained programmable heat transfer behaviors over time, are still unexplored.

In fact, research outside of HCI has shown techniques for related purposes. For instance, material science researchers developed nanoparticle suspensions, which are capable of changing their thermal conductivity responding to external electrical fields [42]. In architecture engineering, researchers also proposed solutions for walls with dynamic infilling fluid to adjust the thermal conductivities of the walls depending on the outdoor temperature variation [7]. Nevertheless, most of these techniques require sophisticated equipment to implement, which limits the utilization within the field of HCI to explore the application of interactive objects with programmable heat transfer behaviors.

In this paper, we introduce *Thermaterial*, a method using fluidic composites with controllable thermal properties to enable temperature behaviors without continuously consuming energy, instead utilizing programmed ambient heat transfer behaviors. We first presented the fluidic composition and fabrication method of *Thermaterial*, which is easy to implement with off-the-shelf equipment. We next evaluated the tunable thermal insulance of *Thermaterial*, revealing changes of up to 3.0 times. We then explored potential applications of objects with programmable ambient heat transfer behaviors, utilizing proof-of-concept prototypes built on *Thermaterial*. These include reconfigurable passive thermal markers, tunable thermal wearables, containers with customized thermal-insulated layouts, and dynamic thermal-insulated enclosures.

2 RELATED WORK

2.1 Techniques of thermal interfaces in HCI

Researchers within HCI fields have developed various techniques to create objects with controlled temperature behaviors for diverse

purposes, such as thermal feedback devices, thermal displays, or tangible interfaces with thermoreactive materials.

Within these works, a majority of them have used electro-thermal elements to programmably selectively heat up or cool down within or on the object, such as using resistive heaters [14, 17, 28, 33, 34] or Peltier elements [18, 19, 31, 36]. With the same goal, researchers have also explored methods of utilizing heat convection from external or internal fluids, such as applying hot air flow to the object with heat guns [16, 27, 35], or pumping fluids with different temperatures inside the object via hydraulic systems [9, 20, 38]. However, all of the above techniques require continuous energy consumption or bulky connected hardware to control the temperature, which partially limits their use in practice.

In contrast, alternative approaches used materials' intrinsic thermal properties and their interaction with ambient heat to produce passive thermal behaviors. For instance, Alex et al. [23] combined different thermally conductive materials on enclosures of devices to alleviate the overheat caused by heat concentration. Zhang et al. [41] employed the Voronoi air-exposure pattern on casts to regulate the convective heat dissipation behavior at different locations for thermal comfort. InfoPrint [12] and ArTuga [5] fabricated markers from materials with different thermal conductivities to achieve distinct temperature display patterns. However, these materials can only provide one predefined set of thermal properties once fabricated. Objects comprising materials with controllable thermal properties, which allow self-contained programmable heat transfer behaviors over time, are still unexplored.

In this work, we introduced a method that uses fluidic composites to control thermal insulance on materials, which further enables objects with programmable ambient heat transfer behaviors.

2.2 Programmable thermal materials beyond HCI

Researchers beyond HCI have investigated different methods to design and develop materials with controllable thermal properties for programmable thermal behaviors.

In material science, researchers explored a range of advanced materials including nanoparticle suspensions, phase changing-materials, or polymers, that can react to external physical fields, such as electromagnetic or temperature fields, to change their thermal conductivity in a programmable manner [42]. These materials have found applications, such as temperature regulation of devices in dramatically varied conditions [6], or metamaterials that performed functional temperature patterns such as a thermal cloak [37]. Using a different approach, research in architecture engineering has explored adjusting the thermal conductivity of building walls by altering fluidic medium inside [7]. This method not only enabled programmable heat transmission behavior through a building's envelope but also proved to be beneficial for energy saving of interior temperature regulation [30]. Nevertheless, most of these techniques require sophisticated equipment to implement, which limits their utilization within the field of HCI to explore the application of interactive objects with programmable heat transfer behaviors.

In this work, we explored the application of objects with programmable heat transfer behaviors in four scenarios, leveraging

prototypes built on materials with controllable thermal properties, which is easier to implement with our method.

3 THERMATERIAL METHOD

Inspired by research in architecture engineering regarding dynamic insulation materials [30], we proposed *Thermaterial*, a fluidic composite with dynamic thermal insulance¹ to program ambient heat transfer behaviors on objects.

3.1 Thermaterial structure

As shown in Figure 2, by altering the fluidic medium within the fluidic channels, we can adjust the overall thermal insulance of the composites accordingly, thus programming the heat transfer effect through the material.

The selection of materials and the dimensions of the structure both influence the overall thermal insulance. In this work, we selected water and air as the embedded fluidic medium due to their significantly different thermal conductivities ($0.6W/(m \cdot K)$ for water and $0.026W/(m \cdot K)$ for air), their widespread availability, and non-toxicity. We used polycarbonate and silicone as substrate materials because they have a maximum operating temperature above $100^{\circ}C$ [1, 2].

We selected the dimensions of the material structure based on a series of considerations, including laminar flow inside the fluidic channels, watertight walls of the fluidic channels, acceptable quality and consistency of selected fabrication techniques, and a minimal overall thickness with notable tunable thermal insulance. Building on prior research, which demonstrated a design of fluidic composites for similar engineering scenarios [38], we optimized this existing design to meet the above considerations through multiple tests. The final setup of the dimensions of a 5mm-thick Thermaterial is illustrated in Figure 2.

Note that, as an exploratory study of this method within HCI, we selected the above specific materials and dimensions as a case study, while acknowledging other optimized choices (e.g., using liquid metal as a fluidic medium) that could potentially enhance performance.

3.2 Fabrication method of Thermaterial

We fabricated Thermaterial by first fabricating the substrate of the fluidic composites, then installing the inside fluidic medium. We introduced methods for fabricating both rigid and flexible substrates.

To fabricate rigid substrate, we used Ultimaker 3 & S5 fused deposition modeling (FDM) 3D printers to print with PolyLite polycarbonate (PC) [2] materials (Figure 3a). We set the infill of printing to 100% and set the printing layer height to 0.1mm. All substrates were printed in a vertical orientation to perform a uniform quality of the fluidic channels. The post-process steps of 3D printing include smoothing the surface with sandpaper, and then applying a minimal layer of UV glue to seal the outer surface of the fluidic channels (Figure 3b-d).

To fabricate flexible substrate, we used Smooth-On SORTA-Clear 12 [1] as material for silicone casting. We first cast the material with one side open using a 3D printed mold (Figure 3e,f), and then

¹In this work, we use *thermal insulance* to describe the material's resistance to heat flow per unit area.

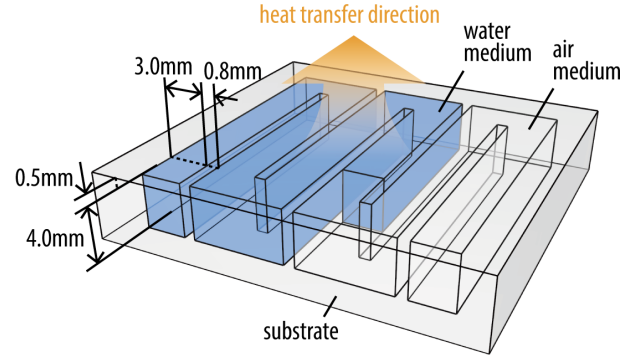


Figure 2: Thermaterial composites, which is a fluidic composite with embedded fluidic channels and performs tunable thermal insulance when changing the infilled fluidic medium.

cast a 0.5mm-thickness silicone membrane on the other side to seal the fluidic channels (Figure 3g,h).

Note that, in this work, the printing files of the substrate and the silicone casting mold were all designed using parametric programs in the Rhino 7 Grasshopper platform, which could be integrated into a complete design tool in the future.

Once the substrate of Thermaterial has been fabricated, we installed the inside fluidic medium with a few accessories, including a syringe with a pre-defined label for applying a certain amount of fluid, a tip, and a silicone tube for connection (Figure 3i). We used vinyl tapes for sealing the inlet of the substrate (Figure 3j).

Note that these accessories are only used when users need to alter the infilled fluidic medium during preparation. These accessories also have the opportunity to be integrated into miniature electrical hydraulic systems [21, 26] for self-contained control in the future. Although this step may require external manual operations or consume energy, we clarify that the benefit of the method in this work is to achieve temperature control without continuously consuming energy (e.g., heat generation of management), instead utilizing programmed ambient heat transfer behaviors by designing the thermal insulance of Thermaterial on the object.

3.3 Tunable thermal insulance of Thermaterial

Based on the Thermaterial structure in Figure 2, theoretically, a lower water-filling ratio will cause a larger thermal insulance of the composites, since air is much less thermally conductive than water. To evaluate this tunable thermal insulance of the composites, we conducted a quantitative experiment on a typical case built on Thermaterial with the rigid substrate.

Figure 4a-b shows the experiment setup, where we used a container with a hot and a cold water tank, separated by a middle wall made of Thermaterial. When water at two different temperatures is added to these two water tanks, the temperature of the two tanks will tend to be consistent due to heat conduction from the Thermaterial middle wall. Thus, we can evaluate the thermal insulance of Thermaterial by measuring the the temperature-neutralizing rates of the two water tanks.

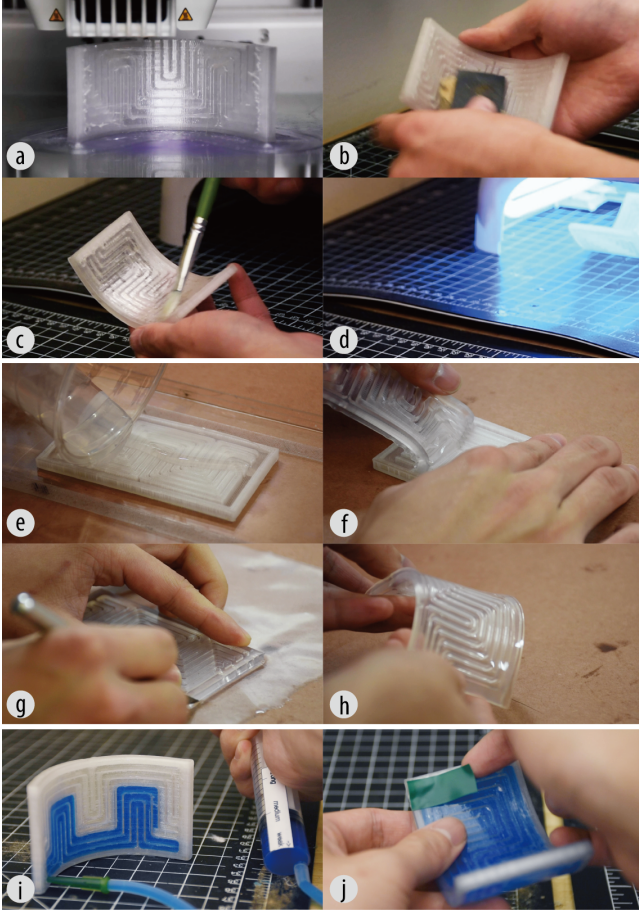


Figure 3: (a-d) Fabricate rigid Thermaterial with FDM 3D printing and post-processing steps. (e-h) Fabricate flexible Thermaterial with silicone casting. (i,j) Install the inside fluidic medium of Thermaterial.

Specifically, we used Equation 1 to approximately describe this temperature-neutralizing process based on the heat conduction law, where $\Delta T/\Delta T_0$ represents the normalized temperature difference of the two water tanks, A denotes the contact area of the water with the Thermaterial middle wall, c and m represent the specific heat capacity and the mass of water from one of the tanks, and r denotes the thermal insulance of Thermaterial.

$$\frac{\Delta T}{\Delta T_0} = e^{-\frac{2A}{cmr}t} \quad (1)$$

During the experiment, we configured the Thermaterial wall with three different water-filling ratios: 100%, 50%, and 0%. Each configuration was tested three times. For each test, we added 90mL of 47°C and 3°C water in the hot and cold tanks as the initial state of the temperature-neutralizing process. We used 10k NTC thermistors as the temperature sensors to measure the temperatures of the two water tanks. We wrapped the container with 2cm-thickness polymeric foam as thermal insulation materials to exclude interference from ambient heat.

The results of the temperature-neutralizing process of the two tanks are shown in Figure 4c, where the normalized temperature difference gradually decreased, however, at a slower rate when configured with a lower water-filling ratio. This phenomenon matched the theoretical analysis. To quantitatively estimate the value of the thermal insulance r of the Thermaterial, we employed the least-square method to fit the experimental data with Equation 1, where $r = 0.0115, 0.0175$, and $0.0345 \text{ Km}^2/\text{W}$ for the water-filling ratio of 100%, 50%, and 0%, respectively. These results showed changes in the thermal insulance of up to 3.0 times, which is comparable with related studies in material science [42]. To estimate the thermal insulance of Thermaterial at other water-filling ratios, we utilized quadratic interpolation based on the experimental data, as shown in Figure 4d.

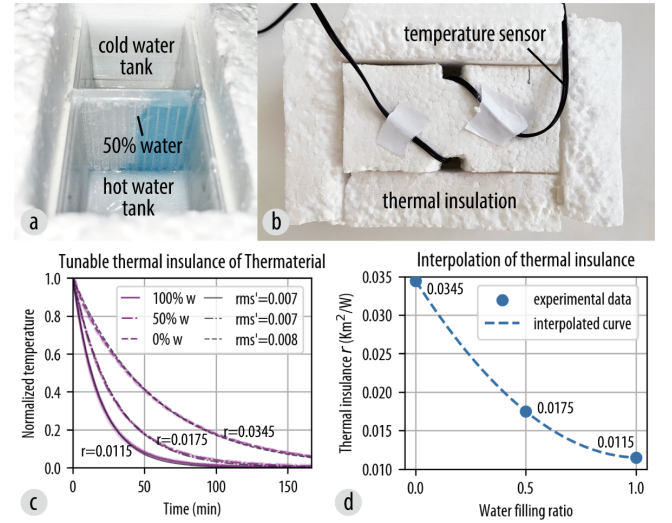


Figure 4: (a,b) Experiment setups to evaluate the tunable insulance of Thermaterial. (c) The normalized temperature difference vs. time at different water-filling ratios, illustrated by the experimental data (translucent magenta lines), the averaged results (opaque magenta lines), and the fitting results (gray lines). (d) Interpolation of thermal insulance at a full range of water-filling ratios.

4 APPLICATIONS AND PROOF-OF-CONCEPT PROTOTYPES

Using Thermaterial, we explored four applications of objects with programmable heat transfer behaviors. We developed proof-of-concept prototypes and conducted quantitative tests for each of these applications.

4.1 Reconfigurable passive thermal marker

We can use Thermaterial to fabricate reconfigurable passive thermal markers as a thermal information display. For instance, Figure 5a,c illustrate two thermal markers infilled with different fluidic patterns of water and air medium. When attached to a heated surface, e.g., a heating machine enclosure, due to the difference in the

thermal insulance of water and air pixels, their surface temperatures will stay different as a result of stable heat transfer. Users can observe distinct thermal patterns through a thermal camera, shown in Figure 5b,d.

These thermal markers have the following advantages:

- (1) The thermal patterns do not require external energy to maintain. The infilled fluid stays stationary once programmed for a desired pattern. During use, the material then programs the natural heat dissipation from the heating surface to the ambient environment to form the thermal pattern.
- (2) These thermal markers can display both visual and thermal information on a heating surface, which may benefit scenarios such as thermal inspection of industrial instruments. As shown in Figure 5e, inspectors can coat the marker with a thin opaque layer containing visual contents, while simultaneously retrieving both temperature and meta-data from multiple thermal patterns through a single thermal camera.
- (3) These thermal markers are reconfigurable. Users can write and re-write the thermal pattern by changing the infilled fluidic pattern. This flexibility enables application on multiple instruments, while each displaying customized information, such as the meta-data of the instrument.
- (4) These thermal markers also have the potential to be used for robot navigation in challenging environments, e.g., smoky or foggy rooms, when attached to heating surfaces in the room. Previous work has shown that infrared light for thermography has a stronger propagation capability compared to visible light in such environments [15].

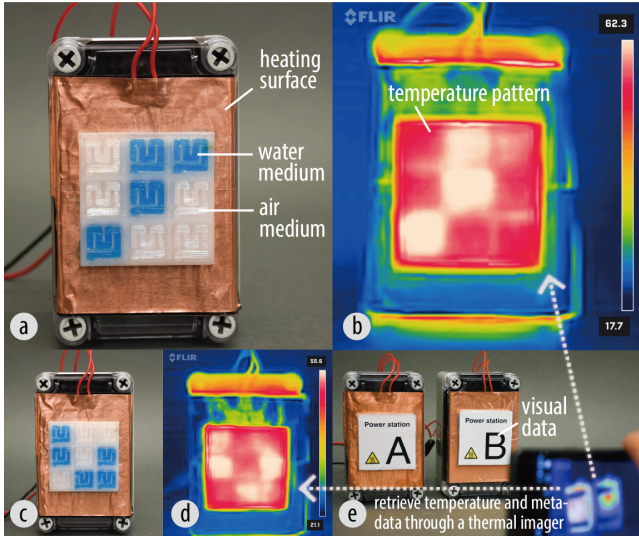


Figure 5: Reconfigurable passive thermal marker. Thermaterial with (a,c) different infilled fluidic patterns and (b,d) distinct surface temperature patterns. (e) Users can retrieve both temperature and meta-data from multiple thermal patterns through a single thermal camera, while simultaneously coating the marker with a thin opaque layer with visual content.

4.2 Tunable thermal wearable

We can use flexible Thermaterial to fabricate tunable thermal wearables with customized skin-contacting temperatures. Figure 6 demonstrates an exemplar ice therapy wearable, where the inner skin-contacting layer is fabricated with a piece of flexible Thermaterial. Users can adjust the cooling skin-contacting temperature by altering the water-filling ratio of the Thermaterial (Figure 6a-c), which serves as a tunable thermal conductive layer between the inside ice packs and the human's skin.

We conducted a preliminary test of the tunable ice therapy wearable on a user's forearm. We installed ice packs with close contact to the Thermaterial layer inside the wearable. The temperature of the ice pack stayed at -1°C during our test, measured by a thermal camera. We then recorded the user's skin temperature after applying the ice therapy wearable with different water-filling ratios on the skin, measured by a 10k NTC thermistor. Figure 6d shows the results, where the skin temperature ranged from 25°C to 30°C , corresponding to 100% to 0% water-filled Thermaterial layer, respectively. Previous work has shown that this skin temperature range can significantly affect the cold sensation on parts of the human body [4].

These results suggest potential application scenarios such as injury therapy or rehabilitation, especially when users require customized cooling temperatures according to personal physical conditions, or need to adjust the coolness over time. Compared to existing methods with similar functions, e.g., a cold therapy machine or wearables combining different thermally conductive materials [23], this method requires no energy supply to maintain the cooling temperature, and it also enables customization of the cooling temperature with only a few accessories. These benefits might be suitable for use in wild environments.

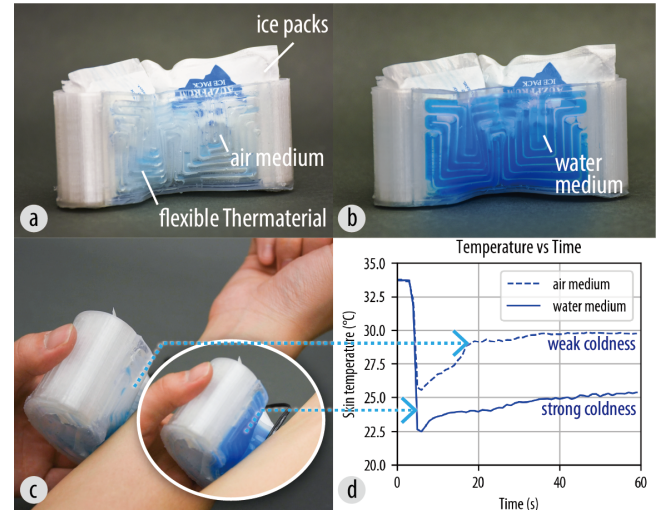


Figure 6: Tunable thermal wearable. (a-c) An ice therapy wearable with flexible skin-contact Thermaterial filled with 0% or 100% water. (d) The cooling skin temperature caused by this ice therapy wearable ranged from 25°C to 30°C .

4.3 Customized thermal-insulated layout in a container

We can use Thermaterial to create containers with customized thermal-insulated layouts. For example, Figure 7 showcases a lunch-box, which contains one left cooling tank, one right heating tank, and two intermediate tanks, where the walls between the intermediate tanks and adjacent cooling and heating tanks are made with Thermaterial. This design allows users to adjust the heating and cooling effects applied to the intermediate tanks through the Thermaterial walls by altering the water-filling pattern.

In our test, we filled the cooling tank with ice and water to maintain a cooling temperature of 0°C , and we attached a temperature-controlled heater to the left wall of the heating tank to maintain of heating temperature of 66°C . We configured the water-filling pattern of the Thermaterial walls as shown in Figure 7a. We then added an equal volume of watery sauce to the two intermediate tanks. After about 25 minutes, as Figure 7b shows, the temperature of the two intermediate tanks stabilized at 31.5°C and 37.7°C , as a result of the heterogeneous stable heat transfer among the heating/cooling tanks and the intermediate tanks.

This example shows the potential to apply a designed internal temperature pattern inside a container with a heterogeneous thermal-insulated layout achieved by Thermaterial separators. Such a method may provide a low-tech approach for preserving temperature-sensitive items, e.g., food or medicine, within different temperature environments inside the same container.

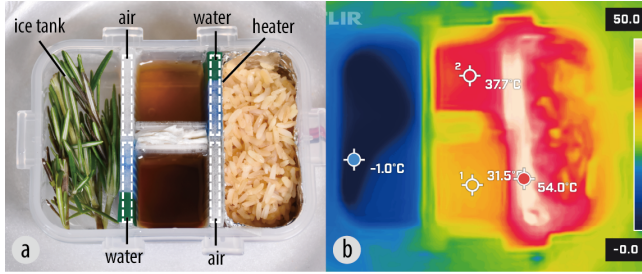


Figure 7: (a) Customized thermal-insulated layout in a lunch-box with Thermaterial as separators. (b) The two intermediate tanks stabilized at two different temperatures as a result of the heterogeneous stable heat transfer between the heating/cooling tanks and the intermediate tanks.

4.4 Dynamic thermal-insulated enclosure

Inspired by previous works in architecture engineering and material science [6, 7], we propose the use of Thermaterial as a dynamic thermal-insulated enclosure for devices requiring dynamic thermal management. For example, Figure 8a,b shows a proof-of-concept prototype of a battery box using Thermaterial as the enclosure. The box contains a silicone heat pad inside, imitating the battery that generates heat. By adjusting the water-filling ratio of the enclosure, the heat dissipation rate of the battery box can be modified.

In one trial of our test, we filled the enclosure with air from beginning to end. We initially set the power of the silicone heat pad at 4.7W and later increased it to 11.3W . During this process,

we observed a 25°C increment in temperature at the inner surface of the ceiling 20 minutes after the power increase. However, in another trial, we first filled the enclosure with the air medium, and then added 100% water inside the enclosure at the moment when increasing the power of the heat pad. As a result, the eventual temperature increment was found to be 5.4°C less than in the former trial. The temperature-changing curves of the two trials are shown in Figure 8c.

These results demonstrated the opportunity to utilize the dynamic thermal-insulated enclosure to mitigate the internal temperature fluctuation in devices experiencing varying inner heating power or fluctuating ambient temperatures. In the future, integrating methods such as miniature hydraulic systems [21, 26] or smart materials, e.g., low-boiling liquids [22, 25], could potentially enable automatically controlled water-filling ratios. This approach may enhance internal temperature management in a more intelligent manner, thus extending equipment adaptability to diverse thermal conditions.

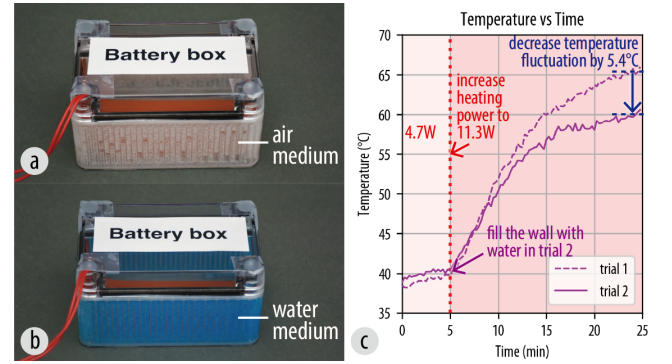


Figure 8: (a,b) A proof-of-concept prototype of a battery box using Thermaterial as the enclosure. (c) The dynamically-insulated battery box demonstrated a 5.4°C less temperature increment when tested with an increasing inner heating power.

5 CONCLUSION AND FUTURE WORK

we introduced *Thermaterial*, a method using fluidic composites with controllable thermal properties to enable temperature behaviors without continuously consuming energy, instead utilizing programmed ambient heat transfer behaviors. We introduced the fluidic composition of Thermaterial with controllable infilling air and water medium, and we presented the fabrication methods for both rigid and flexible Thermaterial. We conducted a technical evaluation of the tunable thermal insulance of Thermaterial, revealing changes of up to 3.0 times. We then explored four applications utilizing proof-of-concept prototypes built on Thermaterial with corresponding evaluation. These included reconfigurable passive thermal markers, tunable thermal wearables, containers with customized thermal-insulated layouts, and dynamic thermal-insulated enclosures.

Future work on Thermaterial includes technical research into its capability to program heat transfer behaviors on objects. For example, how accurately will Thermaterial program the heat transfer

behaviors? What are the methods to computationally model and control the programmability? Can we define a series of high-level programmable thermal behaviors based on the programmability, e.g., programmable surface temperature? Can we develop a design tool to assist users in designing and simulating these programmable thermal interfaces? From a design perspective, can we use Ther-material as a probe to further explore the significance of designing heat transfer behaviors on everyday objects? If allowing using materials with a larger working temperature range, can we expand the application scenarios to larger groups of objects that require heat management, e.g., cooking machines, engine machines, or spacecraft?

These research directions offer intriguing possibilities. We anticipate that our research could provide valuable insights for designing thermal behaviors across various applications.

REFERENCES

- [1] [n. d.]. SORTA-Clear™ 12 Product Information. <https://www.smooth-on.com/products/sorta-clear-12/>
- [2] 2023. PolyLite™ PC. <https://polymaker.com/product/polylite-pc/>
- [3] Byoungkwon An, Ye Tao, Jianzhe Gu, Tingyu Cheng, Xiang 'Anthony' Chen, Xiaoxiao Zhang, Wei Zhao, Youngwook Do, Shigeo Takahashi, Hsiang-Yun Wu, Teng Zhang, and Lining Yao. 2018. Thermorph: Democratizing 4D Printing of Self-Folding Materials and Interfaces. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, Montreal QC Canada, 1–12. <https://doi.org/10.1145/3173574.3173834>
- [4] Edward Arens, Hui Zhang, and Charlie Huizenga. 2006. Partial- and whole-body thermal sensation and comfort—Part II: Non-uniform environmental conditions. *Journal of Thermal Biology* 31, 1 (Jan. 2006), 60–66. <https://doi.org/10.1016/j.jtherbio.2005.11.027>
- [5] Rafael Marques Claro, Diogo Brandão Silva, and Andry Maykol Pinto. 2023. ArTuga: A novel multimodal fiducial marker for aerial robotics. *Robotics and Autonomous Systems* 163 (May 2023), 104398. <https://doi.org/10.1016/j.robot.2023.104398>
- [6] Tingting Du, Zixin Xiong, Luis Delgado, Weizhi Liao, Joseph Peoples, Rajath Kantharaj, Prabudhya Roy Chowdhury, Amy Marconnet, and Xiulin Ruan. 2021. Wide range continuously tunable and fast thermal switching based on compressible graphene composite foams. *Nature Communications* 12, 1 (Aug. 2021), 4915. <https://doi.org/10.1038/s41467-021-25083-8> Number: 1 Publisher: Nature Publishing Group.
- [7] Mohammad Fawaiar and Balázs Bokor. 2022. Dynamic insulation systems of building envelopes: A review. *Energy and Buildings* 270 (Sept. 2022), 112268. <https://doi.org/10.1016/j.enbuild.2022.112268>
- [8] Sebastian Günther, Florian Müller, Dominik Schön, Omar Elmoghazy, Max Mühlhäuser, and Martin Schmitz. 2020. Terminator: Understanding the Interdependency of Visual and On-Body Thermal Feedback in Virtual Reality. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–14. <https://doi.org/10.1145/3313831.3376195>
- [9] Teng Han, Fraser Anderson, Pourang Irani, and Tovi Grossman. 2018. HydroRing: Supporting Mixed Reality Haptics Using Liquid Flow. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology (UIST '18)*. Association for Computing Machinery, New York, NY, USA, 913–925. <https://doi.org/10.1145/3242587.3242667> event-place: Berlin, Germany.
- [10] Alexandra Ion, Robert Kovacs, Oliver S. Schneider, Pedro Lopes, and Patrick Baudisch. 2018. Metamaterial Textures. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, Montreal QC Canada, 1–12. <https://doi.org/10.1145/3173574.3173910>
- [11] Hiroshi Ishii, Dávid Lakatos, Leonardo Bonanni, and Jean-Baptiste Labrune. 2012. Radical atoms: beyond tangible bits, toward transformable materials. *Interactions* 19, 1 (Jan. 2012), 38–51. <https://doi.org/10.1145/2065327.2065337>
- [12] Weiwei Jiang, Chaofan Wang, Zhanna Sarsenbayeva, Andrew Irlitti, Jarrod Knibbe, Tilman Dingler, Jorge Goncalves, and Vassilis Kostakos. 2021. InfoPrint: Embedding Information into 3D Printed Objects. <https://doi.org/10.48550/arXiv.2112.00189> arXiv:2112.00189 [cs].
- [13] Yuhua Jin, Isabel Qamar, Michael Wessely, Aradhana Adhikari, Katarina Bulovic, Parinya Pumpongsanon, and Stefanie Mueller. 2019. Photo-Chromeleon: Reprogrammable Multi-Color Textures Using Photochromic Dyes. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology*. ACM, New Orleans LA USA, 701–712. <https://doi.org/10.1145/3332165.3347905>
- [14] Hsin-Liu (Cindy) Kao, Manisha Mohan, Chris Schmandt, Joseph A. Paradiso, and Katia Vega. 2016. ChromoSkin: Towards Interactive Cosmetics Using Thermochromic Pigments. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA '16)*. Association for Computing Machinery, New York, NY, USA, 3703–3706. <https://doi.org/10.1145/2851581.2890270>
- [15] Shehryar Khattak, Christos Papachristos, and Kostas Alexis. 2018. Marker Based Thermal-Inertial Localization for Aerial Robots in Obscure Filled Environments. In *Advances in Visual Computing (Lecture Notes in Computer Science)*, George Bebis, Richard Boyle, Bahram Parvin, Darko Koracin, Matt Turek, Srikumar Ramalingam, Kai Xu, Stephen Lin, Bilal Alsallakh, Jing Yang, Eduardo Cuervo, and Jonathan Ventura (Eds.). Springer International Publishing, Cham, 565–575. https://doi.org/10.1007/978-3-030-03801-4_49
- [16] Donghyeon Ko, Jee Bin Yim, Yujin Lee, Jaehoon Pyun, and Woohun Lee. 2021. Designing Metamaterial Cells to Enrich Thermoforming 3D Printed Object for Post-Print Modification. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3411764.3445229> event-place: Yokohama, Japan.
- [17] Sven Kratz and Anthony Dunnigan. 2016. ThermoTouch: Design of a High Dynamic Temperature Range Thermal Haptic Display. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA '16)*. Association for Computing Machinery, New York, NY, USA, 1577–1582. <https://doi.org/10.1145/2851581.2892554>
- [18] Kumiko Kushiya, Tetsuaki Baba, Kouki Doi, and Shinji Sasada. 2010. Temperature design display device to use peltier elements and liquid crystal thermograph sheet: "Thermo-Pict neo". In *ACM SIGGRAPH 2010 Posters (SIGGRAPH '10)*. Association for Computing Machinery, New York, NY, USA, 1. <https://doi.org/10.1145/1836845.1836890>
- [19] Guan hong Liu, Tianyu Yu, Zhihao Yao, Haiqing Xu, Yunyi Zhang, Xuhai Xu, Xiaomeng Xu, Mingyue Gao, Qirui Sun, Tingliang Zhang, and Haipeng Mi. 2022. ViviPaint: Creating Dynamic Painting with a Thermochromic Toolkit. *Multimodal Technologies and Interaction* 6, 8 (Aug. 2022), 63. <https://doi.org/10.3390/mti6080063> Number: 8 Publisher: Multidisciplinary Digital Publishing Institute.
- [20] Yuhu Liu, Satoshi Nishikawa, Young ah Seong, Ryuma Niiyama, and Yasuo Kuniyoshi. 2021. ThermoCaress: A Wearable Haptic Device with Illusory Moving Thermal Stimulation. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3411764.3445777> event-place: Yokohama, Japan.
- [21] Jasmine Lu, Ziwei Liu, Jas Brooks, and Pedro Lopes. 2021. Chemical Haptics: Rendering Haptic Sensations via Topical Stimulants. In *The 34th Annual ACM Symposium on User Interface Software and Technology*. ACM, Virtual Event USA, 239–257. <https://doi.org/10.1145/3472749.3474747>
- [22] Qiuyu Lu, Tianyu Yu, Semina Yi, Yuran Ding, Haipeng Mi, and Lining Yao. 2023. Sustainable: Harvesting, Storing and Utilizing Ambient Energy for Pneumatic Morphing Interfaces. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*. Association for Computing Machinery, New York, NY, USA, 1–20. <https://doi.org/10.1145/3586183.3606721>
- [23] Alex Mazursky, Borui Li, Shan-Yuan Teng, Daria Shifrina, Joyce E Passananti, Svitlana Midianko, and Pedro Lopes. 2023. ThermalRouter: Enabling Users to Design Thermally-Sound Devices. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*. Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3586183.3606747>
- [24] Kongpyung (Justin) Moon, Haeun Lee, Jeeun Kim, and Andrea Bianchi. 2022. ShrinkCells: Localized and Sequential Shape-Changing Actuation of 3D-Printed Objects via Selective Heating. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22)*. Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3526113.3545670>
- [25] Takafumi Morita, Ziyuan Jiang, Kanon Aoyama, Ayato Minamino, Yu Kuwajima, Naoki Hosoya, Shingo Maeda, and Yasuaki Kakehi. 2023. InflatableMod: Untethered and Reconfigurable Inflatable Modules for Tabletop-sized Pneumatic Physical Interfaces. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–15. <https://doi.org/10.1145/3544548.3581353>
- [26] Takafumi Morita, Yu Kuwajima, Ayato Minamino, Shingo Maeda, and Yasuaki Kakehi. 2022. HydroMod: Constructive Modules for Prototyping Hydraulic Physical Interfaces. In *CHI Conference on Human Factors in Computing Systems*. ACM, New Orleans LA USA, 1–14. <https://doi.org/10.1145/3491102.3502096>
- [27] Stefanie Mueller, Anna Seufert, Huaishu Peng, Robert Kovacs, Kevin Reuss, François Guimbretière, and Patrick Baudisch. 2019. FormFab: Continuous Interactive Fabrication. In *Proceedings of the Thirteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '19)*. Association for Computing Machinery, New York, NY, USA, 315–323. <https://doi.org/10.1145/3294109.3295620> event-place: Tempe, Arizona, USA.

- [28] Sachith Muthukumarana, Moritz Alexander Messerschmidt, Denys J.C. Matthies, Jürgen Steimle, Philipp M. Scholl, and Suranga Nanayakkara. 2021. ClothTiles: A Prototyping Platform to Fabricate Customized Actuators on Clothing Using 3D Printing and Shape-Memory Alloys. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3411764.3445613> event-place: Yokohama, Japan.
- [29] Martin Nisser, Yashaswini Makaram, Lucian Covarrubias, Amadou Yaye Bah, Faraz Faruqi, Ryo Suzuki, and Stefanie Mueller. 2022. Mixels: Fabricating Interfaces using Programmable Magnetic Pixels. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22)*. Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3526113.3545698>
- [30] Benjamin Park, Wil V. Srubar, and Moncef Krarti. 2015. Energy performance analysis of variable thermal resistance envelopes in residential buildings. *Energy and Buildings* 103 (Sept. 2015), 317–325. <https://doi.org/10.1016/j.enbuild.2015.06.061>
- [31] Roshan Lalitha Peiris, Yuan-Ling Feng, Liwei Chan, and Kouta Minamizawa. 2019. ThermalBracelet: Exploring Thermal Haptic Feedback Around the Wrist. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, Glasgow Scotland Uk, 1–11. <https://doi.org/10.1145/3290605.3300400>
- [32] Roshan Lalitha Peiris, Wei Peng, Zikun Chen, Liwei Chan, and Kouta Minamizawa. 2017. ThermoVR: Exploring Integrated Thermal Haptic Feedback with Head Mounted Displays. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17)*. Association for Computing Machinery, New York, NY, USA, 5452–5456. <https://doi.org/10.1145/3025453.3025824>
- [33] Yasaman Tahouni, Isabel P. S. Qamar, and Stefanie Mueller. 2020. NURBSforms: A Modular Shape-Changing Interface for Prototyping Curved Surfaces. In *Proceedings of the Fourteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '20)*. Association for Computing Machinery, New York, NY, USA, 403–409. <https://doi.org/10.1145/3374920.3374927> event-place: Sydney NSW, Australia.
- [34] Cesar Torres, Jessica Chang, Advaita Patel, and Eric Paulos. 2019. Phosphenes: Crafting Resistive Heaters within Thermoreactive Composites. In *Proceedings of the 2019 on Designing Interactive Systems Conference*. ACM, San Diego CA USA, 907–919. <https://doi.org/10.1145/3322276.3322375>
- [35] Guanyun Wang, Yue Yang, Mengyan Guo, Kuangqi Zhu, Zihan Yan, Qiang Cui, Zihong Zhou, Junzhe Ji, Jiaji Li, Danli Luo, Deying Pan, Yitao Fan, Teng Han, Ye Tao, and Lingyun Sun. 2023. ThermoFit: Thermoforming Smart Orthoses via Metamaterial Structures for Body-Fitting and Component-Adjusting. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 1 (2023), 31:1–31:27. <https://doi.org/10.1145/3580806>
- [36] Graham Wilson and Stephen A. Brewster. 2017. Multi-moji: Combining Thermal, Vibrotactile & Visual Stimuli to Expand the Affective Range of Feedback. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. ACM, Denver Colorado USA, 1743–1755. <https://doi.org/10.1145/3025453.3025614>
- [37] Shuai Yang, Jun Wang, Gaole Dai, Fubao Yang, and Jiping Huang. 2021. Controlling macroscopic heat transfer with thermal metamaterials: Theory, experiment and application. *Physics Reports* 908 (May 2021), 1–65. <https://doi.org/10.1016/j.physrep.2020.12.006>
- [38] Tianyu Yu, Weiye Xu, Haiqing Xu, Guanhong Liu, Chang Liu, Guanyun Wang, and Haipeng Mi. 2023. Thermotion: Design and Fabrication of Thermofluidic Composites for Animation Effects on Object Surfaces. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–19. <https://doi.org/10.1145/3544548.3580743>
- [39] Jiani Zeng, Honghao Deng, Yunyi Zhu, Michael Wessely, Axel Kilian, and Stefanie Mueller. 2021. Lenticular Objects: 3D Printed Objects with Lenticular Lens Surfaces That Can Change their Appearance Depending on the Viewpoint. In *The 34th Annual ACM Symposium on User Interface Software and Technology (UIST '21)*. Association for Computing Machinery, New York, NY, USA, 1184–1196. <https://doi.org/10.1145/3472749.3474815>
- [40] Tengxiang Zhang, Xin Zeng, Yinshuai Zhang, Ke Sun, Yuntao Wang, and Yiqiang Chen. 2020. ThermalRing: Gesture and Tag Inputs Enabled by a Thermal Imaging Smart Ring. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376323>
- [41] Xiaoting Zhang, Guoxin Fang, Chengkai Dai, Jouke Verlinden, Jun Wu, Emily Whiting, and Charlie C.L. Wang. 2017. Thermal-Comfort Design of Personalized Casts. In *Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology (UIST '17)*. Association for Computing Machinery, New York, NY, USA, 243–254. <https://doi.org/10.1145/3126594.3126600>
- [42] ZiTong Zhang and BingYang Cao. 2022. Thermal smart materials with tunable thermal conductivity: Mechanisms, materials, and applications. *Science China Physics, Mechanics & Astronomy* 65, 11 (Sept. 2022), 117003. <https://doi.org/10.1007/s11433-022-1925-2>